Google Summer of Code Proposal

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Idea Name: Physics Guided Machine Learning on Real Lensing Images

Learning the Lens: A Physics-Informed Autoencoder for Dark Matter Substructure Classification

This project introduces a physics-informed deep learning architecture for analyzing strong gravitational lensing images. The model integrates gravitational lensing equations directly into the neural network pipeline through a differentiable lensing simulator.

By jointly performing image classification and lens parameter inference, the architecture produces predictions that are both accurate and physically interpretable. The framework aims to provide a scalable approach for combining astrophysical simulations with deep learning models.

The resulting tools will make a direct contribution to the **DeepLense ecosystem** by enabling researchers to automatically classify dark matter substructures in gravitational lensing images while simultaneously inferring the physical parameters of the underlying lens model. As next-generation surveys such as the Rubin Observatory LSST and Euclid are expected to produce millions of strong lensing candidates, scalable and interpretable analysis pipelines will be essential. By combining classification with physics-based parameter inference, this framework provides both predictive capability and interpretable insights, helping researchers move beyond black-box models toward physically grounded understanding of strong gravitational lensing systems.

1 Abstract

Strong gravitational lensing provides a powerful probe for detecting dark matter substructures within galaxies. Traditional machine learning approaches have demonstrated promising results in classifying simulated lensing images, but they often lack interpretability and fail to incorporate known physical constraints governing the lensing process.

This project proposes a physics-informed deep learning framework that integrates gravitational lensing equations directly into the neural network architecture. A pretrained convolutional encoder extracts high-level features from lensed images, which are used both for classification and for predicting physical lensing parameters such as the Einstein radius and lens center. These predicted parameters are passed through a differentiable gravitational lensing simulator based on the Singular Isothermal Sphere (SIS) model to reconstruct the source and reproduce the observed lensed image.

By jointly optimizing classification and physics-based reconstruction objectives, the model learns representations that are both discriminative and physically meaningful. Preliminary experiments show that the proposed architecture achieves classification performance comparable to strong baseline models while additionally providing interpretable physical parameters and reconstructing the lensing process. This framework also provides a scalable foundation for incorporating more realistic lensing models in future work.

2 About Me

I am an undergraduate Computer Science student at the Indian Institute of Technology (IIT) Palakkad with strong interests in **scientific machine learning, physics-informed neural networks, and computational astrophysics**. My academic background includes coursework in linear algebra, multivariable calculus, probability and statistics, optimization, and classical mechanics.

I have experience developing machine learning systems using **Python and PyTorch**, and I actively work on projects that combine deep learning with physical modeling. For the DeepLense evaluation task, I developed a physics-informed neural network architecture that incorporates gravitational lensing equations into the learning pipeline through a differentiable Singular Isothermal Sphere (SIS) lensing simulator.

In addition to machine learning projects, I am currently involved in research at IIT Palakkad on distributed ensemble learning methods for decentralized datasets. This work involves studying theoretical model risk bounds and building large-scale experimental pipelines in Python.

I regularly practice competitive programming, which has strengthened my algorithmic problem-solving and implementation skills. My GitHub profile contains several projects related to machine learning experimentation, scientific computing, and quantitative modeling.

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Code Repository: https://github.com/Abhiraj212024/deeplens_PINN

3 Approaching the Problem

3.1 Problem Overview

Strong gravitational lensing provides a powerful observational probe for detecting dark matter substructures through distortions in galaxy images. Recent machine learning approaches have demonstrated promising results in classifying simulated lensing images. However, most existing models treat the problem as a purely data-driven classification task and operate as black-box predictors without incorporating the known physical processes governing gravitational lensing.

To address this limitation, this project proposes a physics-informed deep learning framework that integrates gravitational lensing equations directly into the learning pipeline. By embedding a differentiable physics module within the network architecture, the model is encouraged to learn representations that are both discriminative for classification and consistent with the underlying lensing physics.

3.2 Baseline Architecture

As a starting point, a pretrained convolutional neural network is used as a feature extractor. In particular, a ResNet18 backbone is employed due to its strong performance and relatively efficient training time on image classification tasks.

The convolutional layers of the network act as an encoder that transforms the input lensed image into a compact latent representation. This baseline model provides a strong reference point for classification performance while leveraging pretrained visual features that capture spatial structures present in gravitational lensing images.

3.3 Latent Representation and Prediction Heads

The latent representation produced by the encoder is passed through a bottleneck layer to obtain a compact feature vector. This latent representation is then used for multiple prediction tasks through separate neural network heads.

The first head performs the primary classification task, predicting the dark matter substructure class associated with the input lensing image.

The second head predicts physically meaningful parameters that describe the gravitational lens system. For the Singular Isothermal Sphere (SIS) lens model used in the dataset, the predicted parameters include the Einstein radius θ_E and the lens center coordinates (c_x, c_y). These parameters characterize the mass distribution of the lens responsible for producing the observed gravitational lensing effect.

In addition, a decoder network reconstructs the unlensed source image from the latent representation. This reconstructed source image represents the original galaxy prior to gravitational lensing.

3.4 Gravitational Lensing Model

In this project, the gravitational lens is modeled using the **Singular Isothermal Sphere (SIS)** model, which provides a simple and widely used approximation for the mass distribution of lensing galaxies.

The gravitational lensing process is described by the lens equation

$$\boldsymbol{\beta} = \boldsymbol{\theta} - \boldsymbol{\alpha}(\boldsymbol{\theta})$$

where $\boldsymbol{\beta}$ denotes the angular position of the source, $\boldsymbol{\theta}$ denotes the position of the observed image, and $\boldsymbol{\alpha}(\boldsymbol{\theta})$ is the deflection angle produced by the lens.

For the SIS model, the deflection angle is given by

$$\boldsymbol{\alpha}(\boldsymbol{\theta}) = \theta_E \frac{\boldsymbol{\theta} - \boldsymbol{c}}{\|\boldsymbol{\theta} - \boldsymbol{c}\|}$$

where θ_E is the Einstein radius and $\boldsymbol{c} = (c_x, c_y)$ represents the center of the lens.

These parameters are predicted by the neural network and used within a differentiable lensing simulator to reproduce the observed lensed image. The architecture is designed to be extensible to richer lens models such as the Singular Isothermal Ellipsoid (SIE) and Navarro-Frenk-White (NFW) profile.

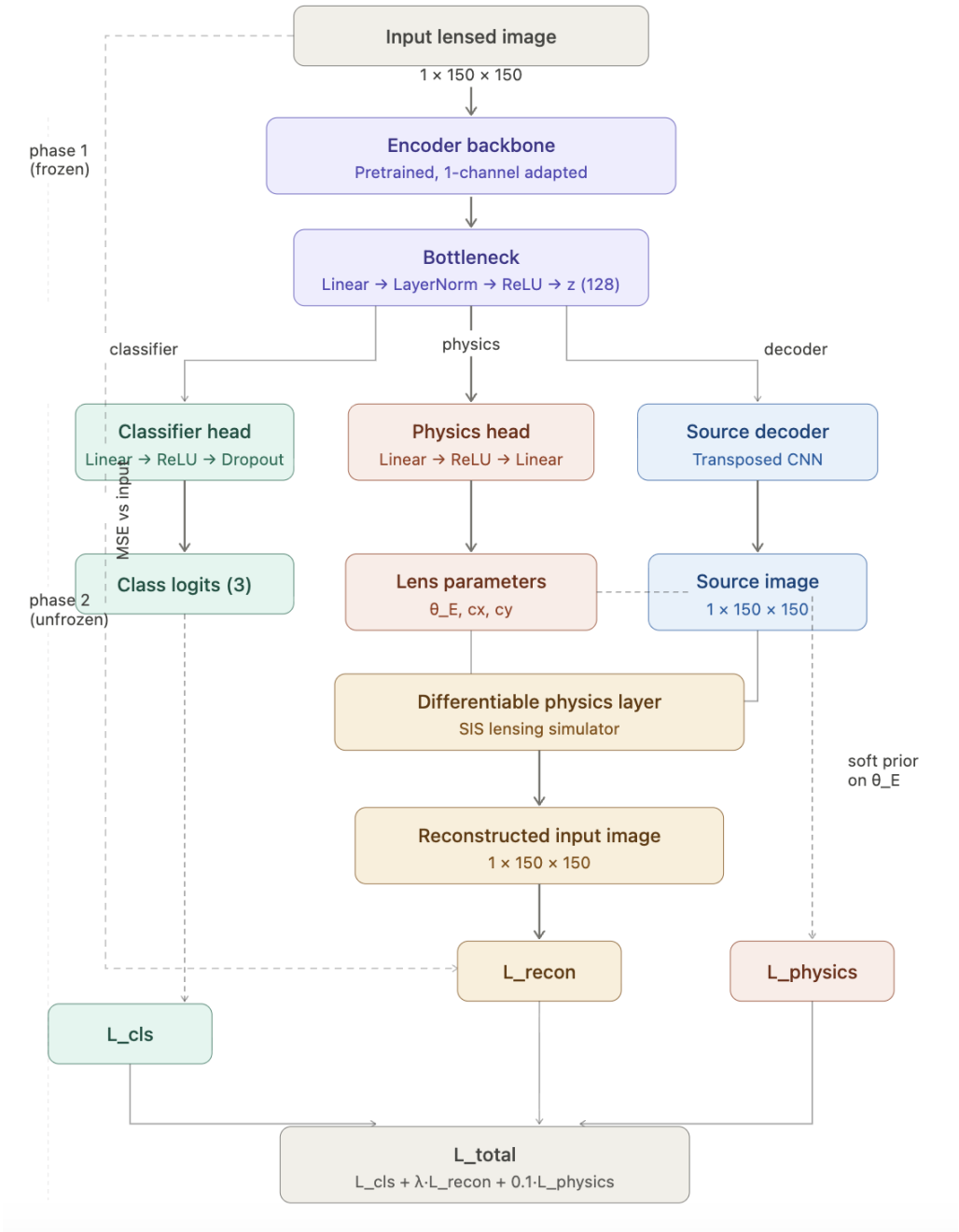

Figure 1: Overview of the proposed physics-informed architecture integrating a neural encoder with a differentiable gravitational lensing simulator.

3.5 Differentiable Lensing Simulation

To incorporate physical constraints into the learning process, the predicted lens parameters are passed through a differentiable gravitational lensing simulator based on the SIS model. The reconstruction pipeline proceeds as follows:

1. A source decoder reconstructs the unlensed source galaxy image from the latent representation.
2. The predicted lens parameters (θ_E, c_x, c_y) are used together with the reconstructed source

image.

3. A differentiable SIS lensing layer simulates the gravitational lensing process to generate a reconstructed lensed image.

The reconstructed lensed image is then compared to the original observed image. By enforcing consistency between the simulated and observed images, the model is encouraged to learn physically meaningful lens parameters. A schematic overview of the complete architecture is shown in Figure 1.

3.6 Multi-Task Learning Objective

The proposed architecture is trained using a multi-task objective that combines classification, reconstruction, and physics-based regularization. The classification task is optimized using cross-entropy loss L_{cls}, reconstruction quality is measured with mean squared error L_{recon}, and physical consistency is enforced through a soft prior on the predicted Einstein radius L_{physics}.

The final training objective combines these components:

$$L_{\text{total}} = L_{\text{cls}} + \lambda_{\text{recon}} L_{\text{recon}} + \lambda_{\text{physics}} L_{\text{physics}}$$

where λ_{recon} and λ_{physics} control the relative importance of each term. In preliminary experiments, λ_{physics} was set to 0.1, while λ_{recon} was gradually increased during training to stabilize optimization.

4 Evaluation Metrics

The proposed model will be evaluated using both machine learning and physics-based metrics to assess classification performance, reconstruction quality, and the estimation of physical parameters.

Classification Performance: Accuracy, F1-score, and ROC-AUC on validation and test sets across all dark matter substructure classes.

Reconstruction Quality: Mean Squared Error (MSE) between reconstructed and observed images; Structural Similarity Index (SSIM) for perceptual similarity.

Physical Parameter Estimation: Prediction error for the Einstein radius θ_E and lens center coordinates (c_x, c_y).

5 Preliminary Results

Preliminary experiments were conducted to evaluate the feasibility of the proposed physics-informed architecture. The goal was to determine whether incorporating physical constraints through the lensing equation could preserve classification performance while enabling the model to learn interpretable physical parameters.

5.1 Baseline Comparison

The physics-informed model achieves a macro ROC-AUC of **0.9900** compared to **0.9948** for the baseline ResNet18. This modest gap reflects an expected tradeoff: by constraining the network to produce physically consistent lens parameters, the model sacrifices a small amount of raw

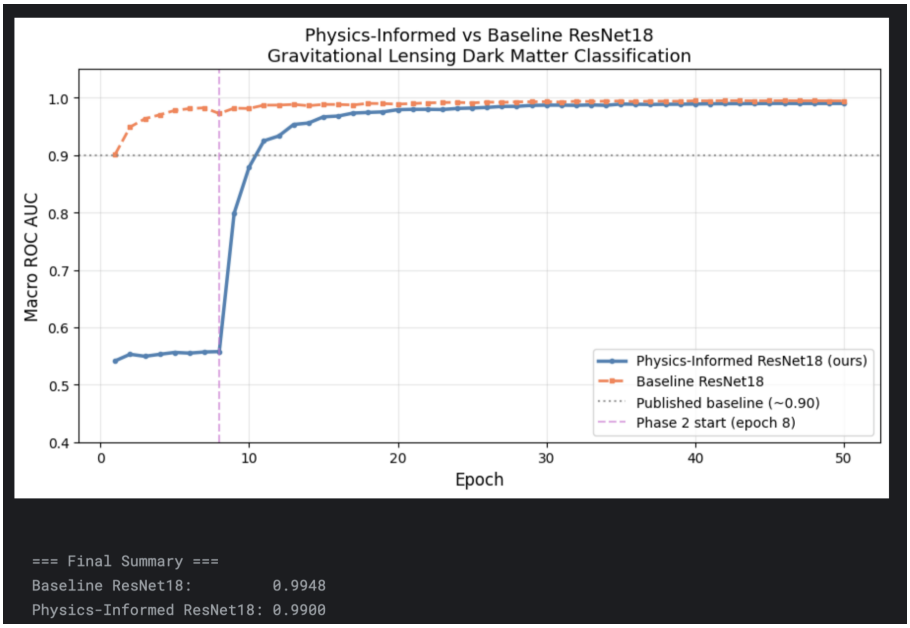

Figure 2: Comparison of classification performance between the baseline ResNet18 model and the proposed physics-informed architecture.

discriminative capacity in exchange for interpretable outputs. Both models comfortably exceed the published baseline of approximately 0.90, confirming that the physics constraint does not meaningfully harm classification performance.

5.2 Reconstruction Quality

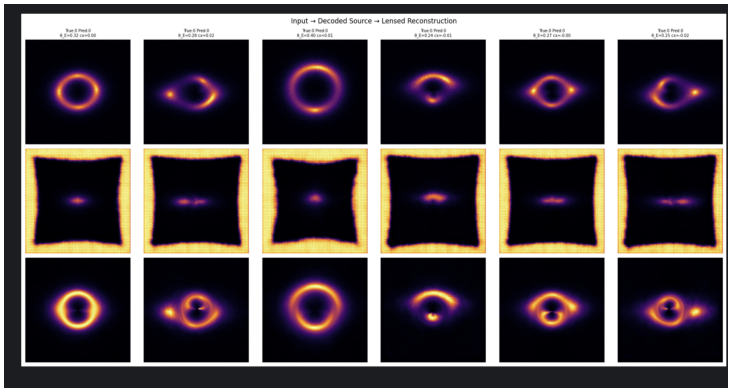

Figure 3: Example reconstructions produced by the physics-informed model. Each triplet shows the input lensed image, the decoded source, and the re-lensed reconstruction.

Reconstructed lensed images generated using the predicted lens parameters closely match the original observations, confirming that the differentiable simulator correctly reflects the governing lensing equations.

5.3 Physics Parameter Inference

The model predicts the Einstein radius θ_E and lens center coordinates across all three dark matter classes. The predicted θ_E values are consistent with the known data generation parameters

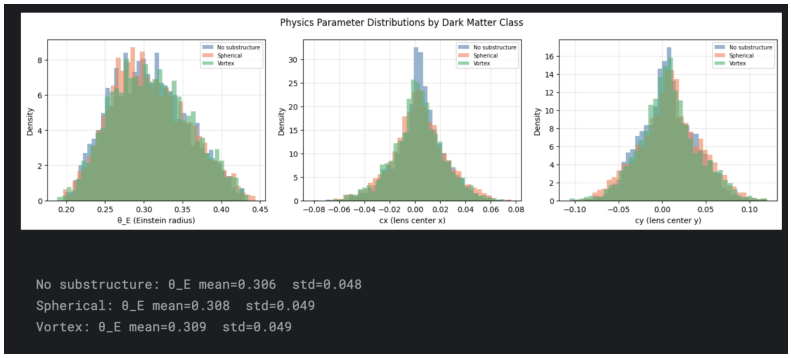

Figure 4: Distribution of predicted lens parameters across dark matter classes. The Einstein radius distributions overlap substantially across classes, as expected for this dataset.

(mean ≈ 0.307, std ≈ 0.049), indicating that the physics head has learned physically plausible estimates rather than arbitrary values.

6 Proposed Timeline

The project will be carried out during the standard 12-week GSoC development period.

Community Bonding Period

During the community bonding period, I will work closely with mentors and contributors to refine the project direction and ensure that the implementation aligns with the goals of the DeepLense project.

- Discuss the current architecture and preliminary experiments with mentors to identify potential improvements and refinements.
- Review the relevant literature on gravitational lensing and physics-informed machine learning to ensure the physical modeling assumptions are appropriate.
- Finalize the lens model(s) to be used in the physics module, including evaluating whether extensions beyond the Singular Isothermal Sphere (SIS) model are required.
- Study the physical principles underlying gravitational lensing to ensure the differentiable simulator correctly reflects the governing equations.
- Identify and organize real-world observational datasets (e.g., HST archival lensing catalogues, JWST early-release data) that may be used to evaluate the model beyond simulated data.

Week 1: Dataset and Physics Setup

- Finalize dataset organization and preprocessing pipeline
- Verify dataset splits for training, validation, and testing
- Finalize the gravitational lensing formulation based on the SIS model
- Identify the lensing parameters to be predicted by the network

Week 2: Architecture Design

- Finalize neural network architecture components
- Design the latent representation used for predicting physical parameters
- Implement differentiable lensing layers based on the SIS equation
- Integrate physics layers with the pretrained encoder

Weeks 3–4: Initial Experiments and Architecture Finalization

- Run initial experiments with the proposed physics-informed architecture
- Compare performance with baseline classification models
- Evaluate reconstruction quality produced by the lensing simulator
- Finalize the architecture to be used for the remainder of the project
- Document architecture design and experiment results

Weeks 5–6: Model Training and Optimization

- Train the finalized model using multi-task learning
- Tune hyperparameters such as reconstruction loss weight
- Analyze training stability and convergence
- Evaluate classification accuracy and reconstruction quality

Midterm Evaluation: At this stage, the physics-informed architecture will be fully implemented and trained, with experimental comparisons against the baseline model.

Weeks 7–8: Extension to Real Data and Anomaly Detection

- Adapt the trained model for anomaly detection tasks
- Explore the model's ability to identify unusual lensing patterns
- Evaluate performance on real-world observational datasets if available

Weeks 9–10: Code Organization and Documentation

- Refactor and organize the codebase
- Prepare reproducible training pipelines
- Write detailed documentation for model architecture and experiments

Weeks 11–12: Buffer Period and Additional Experiments

- Conduct additional experiments and improvements, if needed
- Address unforeseen technical challenges
- Improve visualization and experiment analysis
- Prepare the final report and project deliverables

7 Potential Challenges and Mitigation

Training Instability: Combining three losses may destabilize optimization. Loss weights will be tuned carefully, with λ_{recon} introduced gradually.

Physics Module Integration: Vectorized GPU operations will be used to keep the differentiable lensing layer efficient within the training pipeline.

Generalization to Real Data: Simulated-to-real gaps will be addressed through domain adaptation techniques and dedicated preprocessing for observational noise.

Model Size and Training Cost: Pretrained ResNet18, lightweight decoder architectures, parameter sharing, and mixed-precision training will keep computation feasible.

8 References

- Ana Diaz Rivero, Cora Dvorkin, *Direct Detection of Dark Matter Substructure in Strong Lens Images with Convolutional Neural Networks*, arXiv:1910.00015.
- Prasenjit Saha et al., *Essentials of strong gravitational lensing*, arXiv:2401.04165.
- Michael W. Toomey et al., *LensPINN: Physics Informed Neural Network for Learning Dark Matter Morphology in Lensing*, NeurIPS 2024.
- Michael W. Toomey et al., *HEAL-PINN: Physics-Informed Swin Transformer for Dark Matter Studies for Sparse Lensing Data*, NeurIPS 2025.